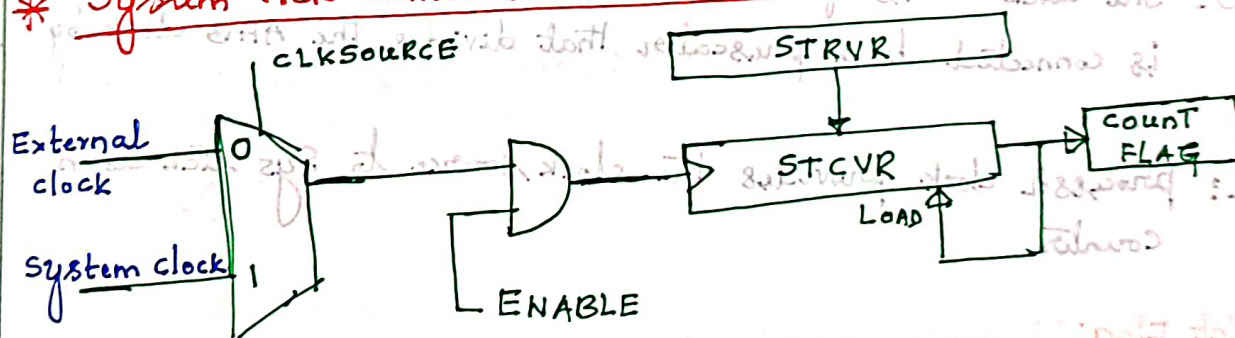


* System Tick Timer:

- Every Arm cortex-M having system tick timer.
- System tick timer allows the operating system to initiate an action on a periodic basis.
- Sys Tick Timer is used widely by operating systems to perform internal operations/actions.
- The SysTick is a 24-bit down counter driven by the processor clock (AHB clock to drive the CPU).
- The System Tick timer counts from an initial value down to 0.
- When timer reaches 0, for the next clock, it underflows and it raises a flag called COUNT and reloads the initial value and starts over again.
- We can set the initial value to any value between $0x000000$ to $0xFFFFFFFF$.

* System Tick Timer Internal structure:



* SysTick Registers: There are three registers in the SysTick module

- 1) STRVR: Sys Tick Reload Value Register
 - 2) STCVR: Sys Tick current value Register
 - 3) STCSR: Sys Tick control and status Register
- The STCVR is a down counter, it receives clock from the system clock or the external clock.
 - The clock is ANDed with enable bit of STCSR, when the enable bit is set then timer starts count decrementing

* Format of STCSR (Sys Tick Control & Status Register) :

D ₃₁	D ₁₇	D ₁₆	D ₁₅	D ₃	D ₂	D ₁	D ₀	
Reserved	COUNT Flag	Reserved	CLK SOURCE	TICK INT	Enable			STCSR

↳ **Enable bit:** Enables/disables the counter

1: When the enable bit is set the timer initializes the STCSR with the value of the STRVR register and counts down until it reaches to zero. Once it reaches zero, for the next clock, it underflows which sets the COUNT Flag to high and the counter reloads the STCSR with the value of the STRVR register and then the process is gets repeated.

0: Disables the counter.

↳ **CLK SOURCE:** provides the choicess of the clock coming from system clock or external clock

0: The clock comes from external clock. The external clock is connected to a prescaler that divides the AHB clock by

8.

1: processor clock provides the clock source to Sys tick down counter.

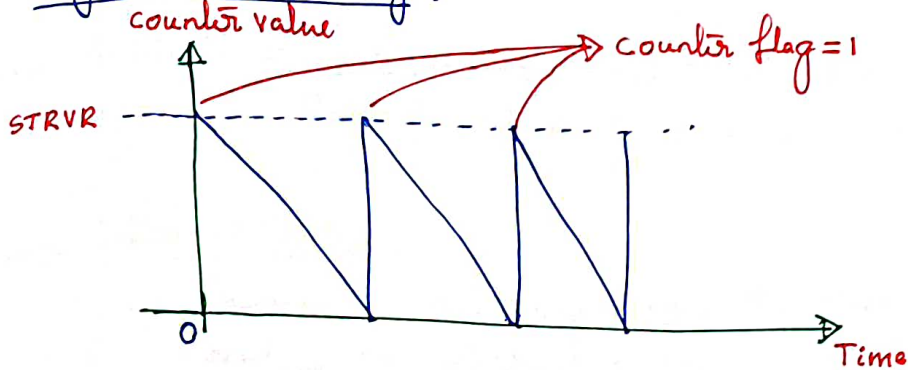
↳ **Count Flag:**

0: The SysTick has not counted down to zero since the last time this bit was read

1: The SysTick has counted down to zero.

* Counter counts down from the initial value and when it reaches 0, the counter is reloaded with the value in STRVR and the count flag is set high. The flag remains high until it is cleared by software.

* System Tick counting :



* NOTE :

- In STM ARM, the PWM (pulse width modulation) is incorporated into the timers.
- The timer/counters can operate as PWM with various duty cycles.
- $TIMxARR$ (Timer Auto-reload Register): This register is used to set the PWM output frequency.
- $TIMxCCRx$ (compare/capture) register: This register is used to set the PWM output pulse width.
- $OCxM$ (output compare mode) register bits are used to generate PWM waves.
- The compare function uses register called $CCRx$ (compare/capture) register.
- The $CCRx$ register content is compared with the value of the counter (CNT Register) continuously. When the values of the CNT register and $CCRx$ register are equal, the corresponding pin performs an action. The action can be set, clear or toggle depending on the configuration.

